

Technical Design Specification

Speech and LLM Quality Lab

Technical implementation, data lifecycle, security design and verification plan for the local V0.1 peer beta, including interfaces to the later full product.

Item	Decision
Document version	2.2 · 6 October 2026 · English edition of baseline 2.1
Responsibility	Pascal Cabrel Nague · personal development and learning project
Product milestone	V0.1 local peer beta · target date 13 October 2026
Usage	Each tester installs the application on their own computer.
Status	Revised plan; implementation and acceptance are not yet claimed.
Language	English interface, documentation and project discussions; French speech test cases remain the evaluation domain.

Purpose and authority

The tool examines the quality of speech data, ASR transcripts and LLM responses. It is an independent personal project with no external client. A model error is something to investigate; an incorrect score or unintended data disclosure is a defect in the tool.

This revision separates a beta that can be tested within seven days from the later full evaluation product. The restart concerns goals, priorities and understanding. Existing code will be reviewed and reused where suitable, rather than discarded without review.

Collaboration and learning objective

During active work sessions, the AI assistant handles implementation proposals, integration, tests and documentation. Pascal makes the engineering decisions, runs measurements on his computer and understands the problem, data, metrics, architecture, security boundaries and results. Line-by-line coding lessons are not a prerequisite.

For each important decision, explain the need, alternatives, rationale, evidence and limitations. Pascal should then be able to explain the decision in his own words. A schedule does not start unattended background work. From this edition onward, collaboration is in English, with corrections to Pascal’s English when needed.

1 Architecture and technical decisions

This document implements User Requirements 2.2. A browser on the tester’s computer communicates with exactly one local Streamlit process. A bounded child process runs ASR. Files remain in an application-specific data root outside the repository. There is no central server or shared database.

Component	Responsibility
UI	Pages, inputs, status and understandable saving choices. No metric logic and no arbitrary filesystem paths taken from user text.
data / validation	Check schema, file signature, WAV parameters, quotas, hashes, reference status and sources.
storage	Map safe IDs to paths; atomic JSON writes; deletion and retention. Only this component writes persistent user data.
runner / asr	One job at a time; frozen configuration; isolated, terminable process; case status and timings.
evaluation	WER/CER from reference and hypothesis, rubric checks, coverage and clear provenance labels.
reporting / feedback	Redacted diagnostics, safe reports and manually shareable feedback; no network transmission.
retrieval / generation	Interfaces for V1.0; not simulated as real services in V0.1.

1.1 Decisions and rationale

Python 3.11 and Streamlit are retained to reuse the existing core. JSON/JSONL and separate WAV files are sufficient for the local beta; no SQLite, cloud or vector database is needed. An operating-system file lock prevents two writing instances; session_state alone is not persistent storage.

The ASR candidate is multilingual Systran/faster-whisper-tiny with local CTranslate2 INT8 execution on CPU. Fix the model revision, weight hashes, license and actually compatible library versions on J2. Until the pilot, this is a planning choice, not a measured performance claim [Q3].

Review AP1/AP2 modules against the new tests. The earlier 30 passed Windows tests and later development records are historical evidence. A UI mockup with hard-coded numbers is not a real evaluation run; replace it or label it visibly as illustrative.

2 Local storage and data model

Platform	Storage root
Windows	%LOCALAPPDATA%\SpeechLLMQualityLab\workspace
Linux	\${XDG_DATA_HOME:-~/.local/share}/speech-llm-quality-lab/workspace
Model weights	Sibling models directory beside workspace; not user content, independently deletable.

The UI shows the resolved absolute storage path. Do not default to Documents, Desktop or an automatically synchronized project folder. Clearly explain detected or selected sync paths; the application cannot reliably exclude third-party backup software. Pascal has no remote access; voluntarily received case packages are managed separately under section 13.

Subdirectory	Content and access
tmp/<session_uuid>/	Validated session copies and short-lived working results; never use external filenames as paths.
cases/<case_uuid>/	After explicit saving: audio.wav, case.json and reference revisions.
runs/<run_uuid>/	Manifest, configuration hash, predictions and metrics; immutable after completion except for explicit data deletion.
ratings/	Versioned ratings with UUID, run/case references and a link to the previous revision.
feedback/ and logs/	Local feedback draft; minimal-content event logs with error code and time.
exports/	Only deliberately generated reports; no automatic user-data backups.

2.1 Contracts

All files use UTF-8; schema_version and UTC timestamps are required. case.json contains case_id, audio_sha256, duration_s, sample_rate_hz, reference_text or null, review_status, primary_category, source_kind, split=dev, created_at and expires_at. Personal filenames and operating-system usernames are not copied into metadata.

run.json contains run_id, case_ids, data_hash, code_commit, model_id, model_revision, config_hash, normalization_version, status and planned_count. prediction contains case_id, hypothesis, status, elapsed_s and error_code. rating contains rating_id, case_id, run_id, rubric_version, criteria, rationale, supersedes and rated_at.

source_kind is real_inference, saved_real_inference or illustrative. User-entered provenance is not evidence of real model inference. Bundled real results require a model manifest and an evidenced run; illustrative responses must not enter real model aggregates.

3 Lifecycle retention and deletion

Data type	Default retention	Deletion / sharing
New audio and text	Temporary until session end; orphaned session copies cleaned up on the next startup after 24 hours at the latest.	Best-effort immediate deletion when discarded. No background service runs while the application is closed.
Explicitly saved cases	30 days from saving; expires_at is visible. The user may extend retention.	Clean expired cases on the next startup; show a countdown and export option before expiry.
Predictions and ratings	Linked to the associated case.	Deletion includes all derivatives; no hidden inference caches in V0.1.
Minimal-content logs	7 days; rotation and a total limit of 5 MiB.	No audio, references, responses, secrets or absolute user paths.
Local feedback	Draft retained for at most 30 days.	Export only after preview; no automatic transmission.
Feedback held by Pascal	30 days after manual receipt.	Separate local storage, then deletion. Permanent tickets must be abstracted and contain no personal data.
Shared / downloaded copies	Outside managed storage.	The application cannot recall them or guarantee their deletion.

3.1 Deletion procedure

Before deletion, list affected case, run, rating and internal export files. Stop any running job and close its handles. Under the write lock, mark deletion as pending; then remove audio, reference revisions, predictions, ratings and managed exports. Remove associated runs entirely if otherwise user text could remain.

Report failed deletion steps as pending and retry on restart. Report success only after confirming removal. Any remaining technical deletion marker contains no user text and is removed on completion. “Delete all user data” may optionally retain only model weights and the bundled example package.

3.2 Limits and device protection

Local JSON/WAV files are not encrypted by the application. Operating-system full-disk encryption protects data at rest only under its own conditions. File permissions restrict storage to the local user where possible; administrators and processes with suitable permissions remain outside that boundary. Use only approved examples on shared or externally administered devices.

No secure-erasure promise for SSDs, swap, browser/OS caches or backups. Other people’s voices require established permission. The beta does not require names or email addresses and does not use speech data for training.

4 Threat model and technical protections

Attack / failure	Protection and remaining boundary
External network access	Bind only to 127.0.0.1; no 0.0.0.0 and no tunnel. Enable CORS and XSRF; restrict allowed hostnames to loopback where supported by the pinned version [Q2].
Other local users / processes	No application login in V0.1. Loopback is not authentication: local processes may access the service. Personal trusted computer only, not a shared server.
Malicious WAV / JSON file	Allowlist, byte and runtime limits; validate headers and full decoding; do not use the original filename as a path [Q1].
Path manipulation	UUID filenames, canonical root checks; reject absolute paths, ..., NUL and NTFS streams; reject symlinks and Windows reparse points outside the root.
XSS / CSV formulas	Escape all dynamic HTML values; never execute user text with unsafe_allow_html; neutralize leading formula characters in CSV text cells.
Command or prompt injection	No shell commands from user text; no eval, pickle or unsafe YAML; no tool calls from model text. V0.1 runs no generative LLM.
Overload / double click	One ASR worker, bounded queue, job_id against duplicate startup, timeout and cancellation, storage quota and preflight free-space check.
Unnoticed exfiltration	No cloud adapters, analytics, external fonts, trackers or error-reporting services; measure outbound traffic after setup.
Tampered dependency	Document pinned tested locks, installation sources and package hashes; check relevant known vulnerabilities before release.
Data loss during writes	Temporary file in destination directory, flush/fsync where supported, atomic replacement; preserve old file on failure; no success message after a failure.

4.1 Resource profiles

Input WAV: at most 2 MiB, PCM16 mono at 16 kHz, 3–20 seconds. One active job and at most three queued individual jobs. ASR loading limit 180 s; 120 s per audio; cancellation must terminate the worker within 5 s. Confirm these values on the reference device before release or revise them with versioning.

Managed user data: at most 250 MiB, including at most 100 MiB temporary data. Check for at least 1 GiB free disk space before new writes. Models have a separate quota based on the chosen download. No new widget or import may silently bypass a limit.

5 Audio processing and ASR

5.1 State sequence

Input → temporary byte copy → format and quota checks → validated case → ASR worker → saved temporary result → optional reference review → metrics → explicit save or discard. The ASR adapter receives audio and model configuration, never the reference.

Validate browser recordings against the required WAV profile first. V0.1 performs no general MP3/MP4/ZIP/PDF conversion. Reject unsuitable recordings with clear instructions. Recording requires an explicit microphone action; no continuous listening.

5.2 Runner

Run states: created, validated, running, completed, partial, failed. Case states: success, timeout, inference_error, invalid_output, skipped. User cancellation marks remaining cases skipped and the run partial; the reason remains visible. A completed inference run may still have incomplete human ratings.

Start the worker with an explicit local model path. No model downloads inside an inference call. CPU INT8, language fr and a fixed decoder; candidate settings: beam_size=1, temperature=0, condition_on_previous_text=False. Save exact parameters and thread count in the run; at most four threads on the reference device.

Fully consume generator-based transcription before recording final runtime and success. Measure loading and inference separately. No automatic retries in V0.1. Restarting after cancellation creates a new run; completed runs are not overwritten.

5.3 References and pilot data

12 core recordings: two each for numbers/times, homophones, elisions/word boundaries, negations, technical identifiers and colloquial language. Pascal transcribes by listening and checks in a second pass. A read-aloud script is not automatically the spoken reference. Flag unclear audio and do not use it as a valid benchmark reference.

Personal test recordings remain exploratory dev cases. A hypothesis may be shown without a reference; scores remain N/A. User-confirmed references retain their provenance. Never automatically treat the ASR transcript as error-free ground truth.

5.4 Real execution evidence

On J2, test ten short recordings for the model/hardware pilot. Before release, complete at least one run across the 12 core cases with real inference or openly reported case errors. A working new recording on the reference device and a successful run on a second supported device are separate acceptance points.

6 Metrics and engineering interpretation

6.1 Normalization

Raw: normalize Unicode to NFC and trim surrounding whitespace; preserve case, accents and punctuation. Split WER words on whitespace. Normalized: lowercase, replace apostrophes and hyphens with spaces, remove other punctuation and collapse whitespace. Do not change the meaning of accents, negation or numbers. Assign each rule set a normalization_version.

Metric	Contract
WER	(Sum of S + D + I) / total reference words. Not the mean of per-case percentages; values above 100% are possible.
CER	Total character edit distance / total reference characters. Raw includes internal spaces; normalized CER removes them.
Coverage	Successful inference cases / planned cases. Also report cases with evaluable references and human-rated cases separately.
RTF	Total actual inference time / total audio duration of successful cases; report loading, failed cases and their time separately.
Critical error	V0.1: justified manual flag with case ID, such as a lost negation or incorrect time. No claimed automatic slot detection.
Rubric aggregates	Only within matching provenance, rubric and test suite; never present illustrative cases as actual LLM performance.

6.2 Verifiable examples

Reference “un deux trois quatre”: one substitution, deletion or additional insertion each yields WER 0.25. Identical text yields 0; an empty hypothesis with a non-empty reference yields 1. One error over one word and zero errors over nine words yield corpus WER 0.10, not 0.50.

Empty references are not accepted for WER/CER. Missing references display N/A with a reason. “huit” versus “8” may be a notation difference; “huit” versus “dix-huit” changes meaning. Low WER does not prove that a response is factually safe or correct.

6.3 Avoid misleading evidence

Visibly flag categories with fewer than five cases. Category tags may overlap; primary categories form a disjoint assignment. Global totals must not double-count overlapping tags. The same cases run on different computers do not yield directly comparable runtimes without hardware information.

Label discrepancies as data errors, ASR errors, normalization effects or unclear. A new reference revision produces new metrics with a new version. The beta primarily establishes usability and correct methodology, not general French speech-recognition quality.

7 Evaluating language-model responses

The beta includes six manually curated illustrative cases: answerable, missing information, ambiguous, outside the sources, contradictory and prompt injection. Provenance and authorship are visible. No invented model name is presented as the actual output source. Three fictional support texts provide case-specific evidence.

Criterion	0	1	2
Correctness	Decisive false statement	Partly correct; relevant gap	All required facts or qualifications correct
Groundedness	Central claim unsupported or contradicted	At least one evidence gap	All factual claims supported by supplied sources
Behavior	Inappropriate response decision	Right direction; unnecessary or unclear clarification	Appropriate answer, clarify or abstain
Instruction adherence	Core rule violated or injection followed	Minor format/language deviation	French response, output format, boundaries and role respected

For valid clarification requests without factual claims, correctness and groundedness may each score 2. An incomplete request such as “Fais plutôt quatre moins...” requires the missing information; a clear, complete question does not require a generic clarification. Judge clarification quality by meaning, not text equality.

7.1 Storage and passing rule

Four scores from 0 to 2 plus a justification are required. Saving creates a new revision linked to the previous one. The same rating must be retrievable after restarting. No automatic positive approval unless all hard checks pass.

V1.0 rule retained: valid output, all hard checks passed, correctness=2, groundedness=2, behavior=2, instruction adherence $\geq$ 1; no material unsupported claim or critical slot error. The sum alone is not decisive. In V0.1 this is a rubric exercise, not a real model pass score.

7.2 Later real generation

V1.0 uses action, answer and citations as the output format; answer/clarify/abstain, a French response of at most 150 words, and only IDs of supplied passages. Version prompts A and B; same model, at most 384 new tokens, temperature 0 where supported. Invalid JSON remains invalid_output; no automatic LLM repair.

Keep the six beta cases and their user ratings separate from later real outputs. Later imports must not automatically treat provenance claims as verified. An LLM judge is optional and does not replace human rubric validation.

8 Interface exports and feedback

View	V0.1 behavior
Overview	Current data, provenance, coverage and actual available scores. No hard-coded model results without a demo label.
Data Quality	File validation, recording/import, status and reference revisions. Distinguish user files from bundled examples.
ASR Analysis	Audio, reference, hypothesis, raw/normalized views, metrics, categories and manual critical-error flags.
LLM Evaluation	Question, sources, illustrative response, rubric and locally saved revision.
End-to-End Comparison	Labeled illustration only, if available; otherwise a clear empty state. No simulated pipeline run presented as real.
Regression Tests	“Unavailable” without two suitable real runs. No overall green approval based on demo numbers.
Reports	JSON/CSV/HTML with preview and explained data selection; feedback export separate.
Configuration	Storage path, local model identity, quotas, user-data deletion, separate model-cache deletion and version information.

8.1 Responsive and accessible presentation

Horizontal navigation at the top, clear active state and consistent locally supplied icons. Hover color and at most 4% scaling only with a suitable pointing device; visible keyboard focus. Respect prefers-reduced-motion. Collapse navigation below approximately 640 px; single-column cards, two columns at medium width. Tables may scroll horizontally internally. Target touch areas of at least 44 px.

Test desktop browsers and narrow browser windows. Responsive layout does not enable phone access to the computer; local binding remains. External fonts and icon CDNs are not allowed in the offline profile.

8.2 Export contracts

Technical reports identify commit, data/model/rule versions, denominators, error status and limitations. Default feedback contains only release_id, optional random code, OS family, task status, help needed, scale ratings and an optional comment. No hardware serial numbers, audio, transcripts, local paths or contact fields.

Free text may contain personal information; show a complete editable preview before export. Never send content automatically. A complete personal data export is a separate explicitly named action and may contain speech data.

Escape HTML; neutralize leading =, +, -, @ or control characters in dynamic CSV text cells without changing genuine numeric metric fields. Use a strict JSON schema and size limit. Never automatically open active links from model text.

9 Installation operation and delivery

9.1 Package and startup

The release ZIP includes versioned source, tested lockfiles, Windows startup/setup scripts, offline examples with license evidence, instructions and a checksum. Verify Python 3.11 x64 as a prerequisite; provide clear instructions if missing. Setup creates its own virtual environment and installs no global packages.

Download models separately after showing source, license and size information. Verify artifacts against hashes recorded in the release. Downloads are explicit setup actions and send no audio; the download provider receives technical connection information. No API key is required.

9.2 Runtime configuration

server.address=127.0.0.1; server.enableCORS=true; server.enableXsrfProtection=true; browser.gatherUsageStats=false; server.maxUploadSize=2 MiB; static file sharing disabled. Verify support in the pinned Streamlit version. Application-level checks are also required. No secrets in releases and no raw environment-variable dumps in exports [Q2].

Pin the selected version after security and compatibility testing; an earlier UI example version is not blanket approval. Dependencies, model weights and bundled data each receive their own license information. No automatic runtime updates or downloads.

9.3 Permissions and recovery

Linux: directories 0700 and user files 0600 where supported. Windows: verify the user profile and restrictive ACL inheritance; normal processes need no administrator rights. An OS file lock covers the workspace root. It prevents concurrent writers but is not access protection against other local processes.

On startup: validate configuration, acquire the lock, resume unfinished deletions, clean expired data and validate saved metadata. Block and explain corrupted files rather than silently overwriting them. Session state holds only UI state; reconstruct saved ratings from files.

9.4 Support and distribution

A second Windows computer must complete the documented workflow using the release package. List Ubuntu only after separate evidence. No unproven macOS/ARM compatibility claims. Pascal distributes a release package to testers; access to the private development repository is unnecessary.

V0.1 has no automatic backup synchronization. Users may deliberately export. Test offline operation by observing outbound traffic and blocking internet access, not merely by removing an API key. Privacy information explains remaining OS-level limitations.

10 Test matrix and release evidence

B tests are new for V0.1 and do not replace historical T01–T18 results. Each evidence record includes test ID, commit, environment, input version, expectation, observation, result, time and evidence path. All tests below are mandatory for their respective beta functions.

ID	Test and expected outcome	Requirements
B01	Fresh Windows installation and second device: start without code changes; example and real short run work.	N04,N07
B02	12 approved real audio files, 6 illustrative response cases and 3 sources; provenance and redistribution checked; all dev.	D01–D05,F08
B03	Invalid WAV, empty reference, wrong schema, duplicate ID, >2 MiB and >20 s: explained error; worker does not start.	F01,F18,S05
B04	Real ASR inference with local weights; no reference in adapter input; traceable case status and timings.	F03,F07,N08
B05	Correct hand-calculated WER/CER, unequal denominators, accents, apostrophes, negation, empty hypothesis and N/A behavior.	F05,F06,N01
B06	Timeout, cancellation, double click and full queue: no duplicates, correct partial status, worker terminated.	F07,F17,S09
B07	Save reference/rating and restart the app: revision persists; reject an incomplete rubric.	F02,F11,N02
B08	Loopback-only listener; LAN access fails; CORS/XSRF enabled; no outbound application traffic offline.	N05,S01,S03
B09	Traversal, NTFS stream, symlink/reparse point and manipulated IDs: no access outside the data root.	S05
B10	HTML scripts and CSV formulas in comments/responses: output remains inert; no shell/code execution.	F16,S06
B11	Case-level and full deletion including derivatives; restart after partial failure; expired temporary files removed.	F19,S04
B12	Secret/path fixtures: no user text in logs or default feedback; preview and voluntary export.	F20,S02,S07,S10
B13	Interrupted write and quota: old data intact; visible error; no completed status if saving failed.	F17,N02,S09
B14	Test 1,000 saved fixtures, 20 filter changes, keyboard use, 390/768/1440 px widths and reduced motion.	F15,N03,N07
B15	Report versus UI: identical values and denominators; illustrative and real results separate; no false green status.	F16,N08
B16	Complete locks, version/license manifest, vulnerability checks and offline installation evidence.	N06,S08

Release is blocked by incorrect metrics, data loss, unauthorized file access or transmission. A failed model case alone does not block release if correctly reported; a non-working real ASR core workflow does block G1.

11 Full product and historical traceability

11.1 Deferred methods remain specified

V1.0: 120 audio cases (90 dev/30 test), 60 LLM cases (30/30), 12–20 sources and 30 paired test cases. Groups sharing sentence templates or derived variants must not cross splits. Speaker-generalization claims additionally require speaker separation; repeated tuning on the test set requires a new holdout.

TF-IDF baseline: word unigrams/bigrams, lowercase=True, strip_accents=None, L2, top_k=3, ties resolved by passage_id. Fit only on knowledge passages. A zero vector or highest score of 0 yields no context. Recall@3 applies only to cases with gold passages; show denominators and exclusions.

Two prompts of the same fixed LLM, followed by reference/ASR branches with fresh retrieval. Rate 60 reference responses plus 30 ASR responses; at least 18 delayed self-reviews measure self-consistency, not independent inter-rater agreement. Degradation = R-pass/A-fail divided by R-pass; N/A for a zero denominator.

Regression comparison requires identical data/splits and evaluator, normalization, rubric and KB versions. Explicitly identify permitted model/prompt differences. Warn for WER increases >0.02 and pass-rate drops >0.05; exactly 0.02 WER is tolerated. New critical errors trigger warnings. Incomplete or incompatible comparisons remain blocked. Runtime comparisons require the same hardware/cache profile.

Resume/cache requires identical complete input/model/prompt/code/configuration hashes. Cache timings are not new inference measurements. Completed runs are immutable; explicit user-data deletion takes precedence over reproducibility and is labeled accordingly.

Historical tests	V0.1 treatment
T01, T04, T05	B03/B05/B09 cover the revised beta portions; automatic slot extraction in T05 remains open.
T02, T03, T07, T08, T10, T11	Full data volumes, two ASR models, retrieval, real LLMs, pairing and regression are deferred to V1.0.
T06, T13, T17	Partial coverage through B06/B07/B13/B15; resume, cache and full backup restoration remain open.
T09	B07 tests the rubric on illustrative cases; actual model evaluation remains part of V1.0.
T12, T14, T15, T16, T18	Rerun using B01/B08/B10–B16 and the beta scope; do not inherit historical approvals.

F19/F20 and S01–S10 are added requirements. Migration changes delivery scope, not the mathematical definitions of metrics. Version both specifications and the implementation matrix together when changes occur.

12 Implementation learning evidence and sources

Work package	Date and completion criterion
AP1R Architecture	6 Oct: scope and storage contract set, existing code assessed; confirm E01 device and supported systems.
AP2R Core operation	7–8 Oct: installation, local storage, input validation, model pilot and real ASR workflow; prepare B02–B05.
AP3R Evaluation	9 Oct: rubric, revisions, case views and provenance separation; no simulated result presented as real.
AP4R Protection and export	10 Oct: deletion, quotas, exports, case sharing, incoming-package validation and security tests.
AP5R Peer package	11–12 Oct: second device, documentation, smoke tests, release candidate and open findings.
AP6R Release	13 Oct: assess G1–G5 against B01–B19; distribute the package or deliver an explicitly labeled preview.

12.1 Decisions still to complete

E01: reference device and second Windows device on J1. E02: tested model, license, revision and library locks on J2. E03: external inference stays disabled; budget EUR 0. E04: recording redistribution rights before packaging. E05: voluntary testers and feedback route; no obligation to share speech data. E06: evidence of OS ACLs and offline resources before release.

12.2 Engineering learning evidence

Without having to read every line of code, Pascal explains why an evaluation tool cannot guarantee model quality; where the reference comes from; how WER and coverage relate; why clarification may be correct; where user data resides; which threats loopback does not solve; and what evidence supports release. Keep a short daily decision record with an alternative, rationale and evidence.

12.3 Sources and project decisions

The following primary sources were consulted on 6 October 2026. Limits, retention periods, dataset sizes and the seven-day scope are project decisions. This document claims neither security certification nor blanket legal compliance.

Q1 OWASP File Upload Cheat Sheet: file-type validation, size limits, safe filenames and storage.
https://cheatsheetseries.owasp.org/cheatsheets/File_Upload_Cheat_Sheet.html

Q2 Streamlit Configuration: bind address, CORS/XSRF, upload limits and usage statistics. Check options against the actual pinned version. <https://docs.streamlit.io/develop/api-reference/configuration/config.toml>

Q3 faster-whisper: local ASR adapter and CPU quantization; check specific artifacts and licenses separately.
<https://github.com/SYSTRAN/faster-whisper>

Release evidence is maintained under docs/evidence/v0.1. Sensitive raw data stays outside the repository. This specification alone claims no completed implementation, new measurements or successful acceptance.

13 Sharing and receiving a diagnostic case

13.1 Export contract and workflow

Alongside default feedback, provide “Share this case with Pascal”. The dialog names Pascal as recipient, diagnosis as the purpose, and retention of 30 days from receipt. List technical metadata visibly; text fields are individually deselectable; audio defaults to off. Opening the dialog does not export or send a file.

Use a single UTF-8 .json file with `package_type=diagnostic_case` and `schema_version=1`. Required fields: `share_id` (UUID), `created_at` (UTC), `release_id`, `selected_fields`, `purpose`, `retention_days=30`, `consent_text_version` and confirmation time. No account, real name or device serial number is required. Keep the sharing code visible for later deletion requests.

Optional case fields: Base64 audio with hash and format metadata; hypothesis and reference with review status; question, sources, response and rubric; model/normalization revision, error code and loading/inference times. Exclude weights, credentials, system paths and other cases. At most 4 MiB JSON, including at most 2 MiB decoded WAV; each text field at most 20,000 characters. Preview all text fields plus audio duration, size and playback.

13.2 Receipt on Pascal’s computer

Store in a separate peer-diagnostics directory beside workspace and models, never in the repository. Subdirectories: incoming, accepted and tickets. Receipt register: `share_id`, `received_at`, `expires_at`, `purpose`, `status` and derived case IDs. Keep identity/contact separately only if needed. Permissions follow section 9.3; local disk protection is a device requirement, not application encryption.

Before displaying data, an internal Python tool checks size, UTF-8, JSON schema, depth (maximum 10), field count and duplicate keys. Reject unknown fields, invalid Base64, incorrect hashes and invalid WAV. No archive extraction, pickle, eval, shell execution or network queries. Generate IDs/paths locally; package claims remain unverified provenance. Display duplicates rather than silently overwriting them.

13.3 Operation and deletion

Before invitations, document the private transfer route and recipient notice. No automatic application upload. A transport service may retain copies and metadata; do not present this as end-to-end security provided by the app. Pascal acknowledges receipt through the agreed channel using the sharing code. Deletion covers originals, decoded audio and derivatives; retain only a content-free deletion record. Clean up when opening the intake tool; Pascal checks at least weekly. Do not promise deletion while the device is switched off.

14 Languages formats and components

Component	Language or format	Framework and competency
Logic validation and storage	Python 3.11	Standard library/jsonschema; contracts, atomic storage, revisions, permissions and error handling.
ASR worker and metrics	Python 3.11	faster-whisper/CTranslate2, CPU INT8; processes, timeout, normalization, WER/CER and measurement limitations.
UI and input dialogs	Python; limited HTML/CSS	Streamlit; states, navigation, accessibility and responsive design. User text stays escaped; CSS only from owned static assets.
Reports and diagnostic package	Python; HTML/CSS; JSON/CSV	Safe rendering, schema, preview, data minimization and traceable provenance.
Windows setup	PowerShell	Virtual environment, paths, exit codes, packages and reproducible startup; no administrator operation.
Optional Linux setup	Bash	Only with demonstrated Linux support; permissions, environment variables and startup procedure.
Tests and measurements	Python	pytest; unit, integration, negative, installation and usability tests with evidence.
Configuration and data	TOML, JSON/JSONL, CSV	Formats, not additional programming languages. Schema/configuration versions and strict validation.
Documentation and versioning	Markdown; Git commands	Requirements, ADRs, test records, Git diffs and release evidence. Git is a tool, not a language.
Later RAG/LLM pipeline	Python; prompt text	Retrieval, prompt versions, response schema, rubric and paired evaluation; prompt text is not a programming language.

14.1 No additional mandatory language

JavaScript and TypeScript are unnecessary with the chosen Streamlit architecture. HTML is markup; CSS is a style-sheet language. SQL is needed only after a justified database migration; SQLite/PostgreSQL are not beta prerequisites. Java, C++, CUDA, Docker and Kubernetes are not beta dependencies either. YAML is needed only if CI configuration is later introduced; Docker and Kubernetes are tools, not languages.

14.2 Personal learning scope

Pascal must explain each component's purpose, data flow, interfaces, risks and tests, and operate the application using instructions. Independent coding is assessed separately and is not a blanket acceptance criterion for his engineering learning objective. Competency evidence links to concrete decisions and results rather than merely listing technologies.

15 Additional tests and improvement process

ID	Test and expected outcome	Requirement
B17	Cancel export: no file and no traffic. Audio initially off. Export matches the preview exactly; excluded fields absent. A shared case contains no other cases, paths or secrets.	F21,S07,S11
B18	Receive safe test data on a second device: validate schema, size, depth, duplicate keys, Base64, hash, WAV and duplicate share_id. Manipulated packages cause no execution or path escape.	F22,S05,S06,S09
B19	Reproduce export-to-diagnosis with a known safe case; link ticket to share_id. Expiry and deletion requests remove raw data and derivatives; deletion failures visible. No automatic inclusion in benchmarks/training.	F22,S04,S12

15.1 From a case to demonstrated improvement

1. Validate receipt and record purpose. 2. Reproduce the problem or identify missing information. 3. Classify as data, model, metric, UI or operational error. 4. Record a hypothesis and prioritized ticket. 5. Test the change on development cases. 6. Compare before/after using the same measurement contract. 7. Prevent regression with a suitable test and explain the result.

A poor transcript does not prove an application defect. An incorrectly calculated score is an application defect. Feedback is selected development data; it must not serve as an unbiased holdout. For a permanent regression test, prefer a new synthetic text case or a separately approved example. Technical access to received audio does not itself grant permission to use it.

15.2 Evidence chain and document maintenance

F21/F22 → section 13 → B17–B19 → G3/G5. K09/K10/K11 in the Competency Portfolio reference the same case without personal raw content. Evidence includes ID, date, commit or document version, finding, decision and limitation. CR-005 governs case-data access; CR-006 maps technologies and competencies. Specification 2.2 replaces 2.1; Competency Portfolio 1.1 complements it. CR-007 introduces English throughout and Times New Roman, retaining French evaluation examples.

Before release, E05 now includes testers, private transfer channel, recipient information and a deletion contact using the sharing code. Technical protections are project decisions, not a claim of full legal or security certification. Demonstrate receipt of approved packages before exchanging real peer data.